



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION

LINUX BASH SCRIPTING LAB

(ENCS356)

Lab file Submitted to

K.R. MANGALAM UNIVERSITY

For

Bachelor Of Technology In

Computer Science Engineering with Specialization

In Cyber Security

Submitted By :

Aman Yadav (2301410010)

Faculty:

Dr. Rajesh Kumar

Experiment No. 1

Exploring Linux System Commands, Run Levels (Targets), and Help Utilities.

Problem Statement

Efficient use of the Linux operating system requires a strong understanding of basic system commands, system states, and built-in help mechanisms. Unlike graphical operating systems, Linux relies heavily on command-line utilities for system interaction, administration, and automation.

The problem is to explore and practice essential Linux system commands, understand system run levels / targets, and effectively use Linux help utilities such as `man`, `--help`, and `auto-` completion to access command documentation and usage information.

Aim

To study and execute basic Linux system commands, understand Linux run levels (systemd targets), and use Linux help utilities such as `man`, `--help`, and tab auto-completion.

Theory

Linux Command Line Interface (CLI)

The Linux Command Line Interface allows users to interact with the operating system by typing commands. It provides better control, flexibility, and efficiency compared to graphical interfaces, especially for system administration and automation.

System Run Levels / Targets

- Modern Linux systems use **systemd**, where traditional run levels are replaced by **targets**.

Common systemd targets:

- `graphical.target` – Multi-user mode with GUI
- `multi-user.target` – Multi-user mode without GUI
- `rescue.target` – Single-user rescue mode
- `poweroff.target` – Shutdown system
- `reboot.target` – Reboot system

Linux Help Utilities

Linux provides built-in help tools for learning commands:

- `man` – Displays complete manual pages
- `--help` – Shows brief command usage
- **Tab auto-completion** – Completes commands and file names automatically

Commands Used

Commands	Description
<code>pwd</code>	Display present working directory
<code>ls</code>	List files and directories
<code>cd</code>	Change directory
<code>whoami</code>	Display current user
<code>date</code>	Show system date and time
<code>uptime</code>	Show system running time
<code>man</code>	Display manual pages
<code>--help</code>	Display help for a command
<code>systemctl</code>	Manage system services and targets
<code>runlevel</code>	Display previous and current run level
<code>clear</code>	Clear terminal screen

Procedure / Steps

Step 1: Login:

Log in to the Linux virtual machine and open the Terminal.

Step 2: Execute Basic Linux Commands in the Terminal.

Commands:

- `pwd` shows current directory
- `ls` lists files
- `whoami` displays logged-in user
- `date` shows system date/time

- `uptime` shows how long system is running

```
(amanyadav@kali-vm)-[~]
$ pwd
/home/amanyadav

(amanyadav@kali-vm)-[~]
$ ls
Desktop  Downloads  LinuxLabTest  Pictures  Templates
Documents  lab        Music        Public    Videos

(amanyadav@kali-vm)-[~]
$ whoami
amanyadav

(amanyadav@kali-vm)-[~]
$ date
Saturday 28 February 2026 05:54:09 PM IST

(amanyadav@kali-vm)-[~]
$ uptime
17:54:12 up 11 min,  1 user,  load average: 0.82, 2.22, 1.51
```

Step 3: Directory Navigation using:

- `cd /home` moves to home directory.
- `cd ~` returns to user's home directory.

```
(amanyadav@kali-vm)-[~]
$ cd /home

(amanyadav@kali-vm)-[/home]
$ cd ~

(amanyadav@kali-vm)-[~]
$ man ls
```

Step 4: Using Manual Pages

- `man ls`
- `man cd`

It Read command details.

- Press `q` to exit the manual.

```
LS(1) User Commands LS(1)
NAME
ls - list directory contents
SYNOPSIS
ls [OPTION]... [FILE]...
DESCRIPTION
List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.
Mandatory arguments to long options are mandatory for short options too.
-a, --all
do not ignore entries starting with .
-A, --almost-all
do not list implied . and ..
--author
with -l, print the author of each file
-b, --escape
print C-style escapes for nongraphic characters
--block-size=SIZE
with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below
-B, --ignore-backups
do not list implied entries ending with ~
-c with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime,
Manual page ls(1) line 1 (press h for help or q to quit)
```

Step 5: Using Help Option:

- `ls --help`
 - `systemctl --help`
- This displays short usage instructions and options.

```
(amanyadav@kali-vm)-[~]
$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.
Mandatory arguments to long options are mandatory for short options too.
-a, --all
do not ignore entries starting with .
-A, --almost-all
do not list implied . and ..
--author
with -l, print the author of each file
-b, --escape
print C-style escapes for nongraphic characters
--block-size=SIZE
with -l, scale sizes by SIZE when printing them;
e.g., '--block-size=M'; see SIZE format below
-B, --ignore-backups
do not list implied entries ending with ~
-C
with -lt: sort by, and show, ctime (time of last
change of file status information);
with -l: show ctime and sort by name;
otherwise: sort by ctime, newest first
-C
list entries by columns
--color[=WHEN]
color the output WHEN; more info below
-d, --directory
list directories themselves, not their contents
-D, --dired
generate output designed for Emacs' dired mode
-f
same as -a -U
-F, --classify[=WHEN]
append indicator (one of */=>@|) to entries WHEN
likewise, except do not append '*'
--file-type
across, horizontal (-x), commas (-m), long (-l),
single-column (-1), verbose (-l), vertical (-C)
--full-time
like -l --time-style=full-iso
-g
like -l, but do not list owner
```

```
(amanyadav@kali-vm)-[~]
$ systemctl --help
systemctl [OPTIONS ...] COMMAND ...

Query or send control commands to the system manager.

Unit Commands:
list-units [PATTERN ...]      List units currently in memory
list-automounts [PATTERN ...] List automount units currently in mem>
                                ordered by path
list-paths [PATTERN ...]      List path units currently in memory,
                                ordered by path
list-sockets [PATTERN ...]    List socket units currently in memory,
                                ordered by address
list-timers [PATTERN ...]     List timer units currently in memory,
                                ordered by next elapse
is-active PATTERN ...         Check whether units are active
is-failed [PATTERN ...]       Check whether units are failed or
                                system is in degraded state
status [PATTERN ... |PID ...] Show runtime status of one or more un>
show [PATTERN ... |JOB ...]   Show properties of one or more
                                units/jobs or the manager

lines 1-20 ... skipping ...
systemctl [OPTIONS ...] COMMAND ...

Query or send control commands to the system manager.

Unit Commands:
list-units [PATTERN ...]      List units currently in memory
list-automounts [PATTERN ...] List automount units currently in memory,
                                ordered by path
list-paths [PATTERN ...]      List path units currently in memory,
```

Step 6: Check System Run Level / Target

- `runlevel`- “Kali Linux uses `systemd`, so traditional `runlevel` commands are deprecated and `systemctl` targets are used instead.”
- `systemctl get-default` shows default system target.

```
(amanyadav@kali-vm)-[~]
$ systemctl get-default
graphical.target

(amanyadav@kali-vm)-[~]
$ |
```

- Use `clear` command to clear the terminal.

Outcomes

- Successful execution of basic Linux commands
- Display of manual pages and help options
- Identification of current run level/target
- Improved command-line efficiency

Result

Thus, basic Linux system commands, system run levels, and help utilities were successfully explored. The student gained practical understanding of Linux command-line interaction and Linux documentation tools.